Chapter 1

INTRODUCTION

1.1 Overview

Modern warfare has transitioned into an increasingly data-driven environment where split-second decisions are critical for survival and mission success. The sheer volume of visual information from surveillance feeds, drones, and tactical environments often overwhelms personnel, leading to cognitive fatigue and a higher margin for error. Manual threat assessment is no longer sufficient to ensure the safety and tactical superiority of ground units in high-stress scenarios. Consequently, there is an urgent need for intelligent, autonomous systems that can process vast amounts of real-time visual data to provide actionable insights and enhance situational awareness.

VeerDrishti is an AI-enabled combat vision system designed to bridge this gap by providing soldiers with a persistent, intelligent observer through a camera-integrated solution. By leveraging state-of-the-art computer vision models and edge computing, the system autonomously identifies enemy combatants, vehicles, and weapons. The primary objective is to offload the mental burden of constant monitoring from the soldier, allowing them to focus on strategic execution. Developed as a lightweight solution suitable for deployment on wearable hardware, the system aims to transform how visual data is utilized on the front lines of national security.

The project incorporates sophisticated methodologies to ensure reliability under diverse and challenging battlefield conditions. It utilizes multimodal image fusion—combining visible and thermal spectrums—to maintain high detection accuracy in low-light or obscured environments. Furthermore, by implementing edge-optimized architectures and advanced feature fusion techniques, the system achieves high-speed inference without compromising precision. This integration of versatile object detection and robust information processing ensures that the system remains a reliable asset, offering a significant technological advantage in modern combat and defense operations.

1.2 Problem Statement

The problem statement is to address the challenges of managing vast real-time visual data in modern battlefield environments, where manual monitoring is slow, error-prone, and inefficient; in order to enhance situational awareness and tactical decision-making, an AI-powered combat vision system is proposed to automatically detect, classify, and assess threats in real time, improving response speed and operational efficiency.

1.3 Motivation

The motivation for VeerDrishti is to minimize casualties and save lives by providing an intelligent, persistent observer in combat environments. Modern warfare involves overwhelming visual data that leads to cognitive fatigue, increasing the risk of fatal human error. The system serves as a technological safeguard, enhancing personnel safety and protecting national security forces during high-stress operations.

1.3.1 Scope

The scope of the project is to develop an autonomous, real-time vision system designed to detect enemy personnel, weapons, and vehicles in tactical environments. The technical implementation will focus on developing a lightweight system, optimized for localized inference on mobile edge devices.

1.3.2 Purpose

The purpose of the project is to provide an autonomous, intelligent observer that delivers real-time situational awareness in high-stress combat environments. By automating threat detection and mitigating cognitive fatigue, the system serves as a technological safeguard dedicated to achieving the project's core mission: minimizing casualties and saving lives.

1.3.3 Applicability

The project will be applicable in active combat operations, border surveillance, and high-risk tactical missions where ground units require instantaneous threat identification. The edge-optimized architecture will ensure reliability in signal-denied zones and low-visibility environments, serving as a critical force multiplier for national security personnel.

Beyond defense, the system's ability to automate hazard detection will be directly transferable to emergency response and disaster management, fulfilling the mandate to minimize casualties and save lives.

1.4 Objectives

The project's primary objectives are to develop a lightweight, real-time combat vision system that can be deployed on wearable hardware, capable of autonomously detecting and classifying threats in dynamic battlefield environments.
Specific objectives for the project include:

i. Implement a lightweight real-time object detection model.

ii. Enable on-screen visual highlighting of identified threats.

iii. Optimize system performance for low latency and smooth operation.

iv. Evaluate accuracy and response time under simulated field conditions.

1.5 Methodology

The project adopts a hybrid deep-learning approach that integrates generative data augmentation with edge-optimized multimodal fusion. Initially, the system utilizes a Generative Adversarial Network (GAN) to synthesize high-quality military training data, addressing scarcity in combat-specific datasets. For real-time execution, the project implements a lightweight detection architecture, such as EfficientDet or YOLOv10, optimized for low-latency performance on edge devices like smartphones or Raspberry Pi. To ensure reliability in varying field conditions, the methodology incorporates multimodal image fusion, specifically merging RGB texture with Infrared (thermal) edge data using a fine-grained region attention mechanism. This ensures robust target classification even during night operations or in cluttered environments. Furthermore, the system leverages a vision-language model (like Florence v2) for zero-shot detection, allowing the identification of ”unseen” or novel threats through natural language supervision. Post-detection, the pipeline filters data and applies an Augmented Reality (AR) overlay to highlight threats instantly, while securing all mission metadata in a local storage layer for subsequent auditing.

1.6 Sustainable Development Goal Relevance

The project aligns with United Nations Sustainable Development Goal 16: Peace, Justice, and Strong Institutions.

Target 16.1: Significantly reduce all forms of violence and related death rates everywhere

VeerDrishti will enhance battlefield situational awareness by detecting and tracking potential threats in real time. This capability will enable defense personnel to respond faster and make more informed decisions during combat situations. By improving awareness and reaction time, the system will help reduce operational risks and potentially decrease casualties.

Target 16.3: Promote the rule of law at the national and international levels and ensure equal access to justice for all

VeerDrishti will record visual data during operations, enabling improved mission analysis and operational review. It will support evidence-based evaluation of decisions made during missions. Such recorded data could also assist investigations related to violations of international humanitarian law, including potential war crimes.

Target 16.A: Strengthen relevant national institutions, including through international cooperation, for building capacity at all levels, in particular in developing countries, to prevent violence and combat terrorism and crime
VeerDrishti will integrate AI-based detection and tracking technologies, strengthening the technological capacity of national security institutions. The project will support peacekeeping operations and the protection of national interests.

1.7 Feasibility Study

The feasibility of the proposed combat vision system is assessed across technical, operational, and economic dimensions.

Technical Feasibility: The proposed system is technically feasible using existing smartphone hardware and computer vision frameworks. Modern phones provide high-resolution cameras, sufficient processing power, and GPU support for real-time object detection. Light weight AI models can be optimized for edge deployment, enabling on-device threat detection without requiring complex external hardware.

Operational Feasibility: The system can be integrated into existing helmet setups with minimal modification, using a mounted smartphone as both camera and display interface. It reduces cognitive load by automating threat detection and visual highlighting. Basic training would enable personnel to interpret AR overlays effectively in dynamic environments.

Economic Feasibility: Using commercially available smartphones significantly lowers development and deployment costs compared to specialized military hardware. Open-source AI frameworks further reduce software expenses. The prototype approach minimizes initial investment, making the system cost-effective for research, testing, and potential scalable improvements in future versions.

1.8 Work Distribution

Name	Work
Gaurav Chauhan	Literature Survey and Report Writing
Gaurav Singh	Literature Survey and Slide Preparation

Table 1.1: Work Distribution Among Team Members

The table 1.1 represents the work distribution between the team members for Major Project (Phase-1).

Chapter 2

LITERATURE REVIEW

2.1 Selected Papers

X. Zhuang, et al. This paper proposes a novel method for military target detection by integrating EfficientDet, a state-of-the-art object detection model, with a Generative Adversarial Network (GAN) to enhance the training data. The authors demonstrate that this approach significantly improves detection accuracy, especially in scenarios with limited training data. The method is evaluated on a military target dataset, showing superior performance compared to traditional detection methods. The authors first use EfficientDet to detect targets in the original dataset. Then, they employ a GAN to generate synthetic images of military targets, which are added to the training set. The enhanced dataset is used to retrain the EfficientDet model, resulting in improved detection performance. The method achieved 92.8% mean Average Precision (mAP) on a custom military dataset, outperforming YOLOv3 and YOLOv5. It maintained real-time performance (15 FPS) on the test hardware. In complex scenarios (dim lighting, occlusions), the model still achieved over 90% detection accuracy. [1]

Ankita, S. Rani, et al. This paper presents an efficient and lightweight deep learning model for recognizing human activities using smartphone sensors. It addresses the limitations of traditional pattern recognition by using automated feature extraction, which is essential for applications in healthcare, elderly monitoring, and sports science. The proposed CNN-LSTM architecture combines convolutional layers for automated feature extraction with LSTM units for processing temporal sequences. The model was trained using the UCI-HAR dataset from Samsung Galaxy S2 smartphones, employing Gaussian standardization to preprocess raw accelerometer and gyroscope signals. The CNN-LSTM model achieved a high recognition accuracy of 97.89%. It proved to be more robust and capable than traditional algorithms, demonstrating that combining spatial feature learning with temporal sequence interpretation effectively identifies activities like walking, sitting, and standing while remaining lightweight for mobile deployment. [2]

V. Kaya, et al. This paper proposes a deep learning model for the automatic detection and classification of seven distinct handheld weapon types. It aims to support security forces by providing a proactive control system that can identify threatening objects in real-time surveillance to prevent criminal activities. Based on the VGGNet architecture, the proposed CNN model uses 25 layers and 337,671 parameters. A new dataset of 5,214 images (assault rifles, pistols, grenades, etc.) was curated from the internet and preprocessed using gray formatting and background cleaning for training via the Keras library. The model achieved 98.40% classification accuracy, significantly outperforming VGG-16 (89.75%), ResNet50

(93.70%), and ResNet-101 (83.33%). It successfully identified weapons with low loss rates (0.052), proving highly effective for real-world deployment on low-cost computers due to its reduced layer complexity and fast training time. [3]

S. D N, et al. A state-of-the-art object detection system for analyzing camera or drone footage. Designed to identify battlefield threats such as vehicles, weapons, and combatants. Uses a vision-language model, combining images and natural language descriptions for better understanding and classification. CLIP links images with text descriptions, enabling detection of new objects without large labeled datasets. SPPELAN Pooling module that ensures fixed feature representation with minimal information loss. N-gram & GLIP Labeler extracts noun phrases and creates pseudo-boundaries for objects to generate training data. K-means Clustering groups similar image regions to improve efficiency in cluttered scenes. Compared with Grounding DINO on the Microsoft COCO dataset, the system showed better efficiency. Lower complexity (48M parameters vs 172M), higher accuracy (AP 35 vs 25.6) and faster framerate (17.6 FPS vs 1.5 FPS), making it suitable for realtime deployment. [4]

Z. Song, et al. EDNet is a deep-learning framework for detecting small objects in UAV/drone imagery. Built on YOLOv10 and optimized for edge computing, allowing efficient use on mobile and low-power devices. The model is scalable, with 7 sizes ranging from Tiny (lightweight) to XL (high performance). C2f-FCA Block uses faster context attention to extract features with lower computational complexity. WIoUv3 Loss Function helps the model focus on high-quality samples and ignore noisy data. Optimized Deployment uses CoreML, TensorRT, and OpenVINO for high-speed performance across different processors. EDNet outperformed models like YOLOv8 and YOLOv10 in accuracy and speed. Higher Accuracy: Up to 5.6% improvement in mAP_{50} compared to YOLOv10 with fewer parameters. Better Efficiency: EDNet-M surpassed YOLOv10-X while using 39.6% fewer parameters. Real-Time Performance: 16–55 FPS on iPhone 12; EDNet-Tiny runs 3 times faster on Raspberry Pi 5. [5]

Z. Shou, et al. This paper evaluates the latest deep learning-based models for 3D human pose estimation (HPE). It explores the evolution from 2D to 3D positioning of articulated joints, highlighting key industrial applications in healthcare telerehabilitation, security surveillance, gaming, and action recognition. Researchers followed the PRISMA protocol to systematically search Scopus, Web of Science, and IEEE databases for articles published between 2014 and 2024. They assessed 30 leading studies based on neural network architectures, benchmark datasets (like Human 3.6M), and metrics like Mean Per Joint Position Error. The review found that while deep learning has significantly advanced 3D HPE, major challenges persist in handling occlusion and real-time processing. Findings emphasize the success of multi-view and sensor-fusion techniques, providing a roadmap for future research toward more robust, generalized models for complex environments. [6]

X. Xiang, et al. A framework for cyber-physical systems such as autonomous vehicles and smart surveillance. It introduces MSAF-Net, a system for human action recognition. It combines multiple visual data types like RGB, depth, and thermal images for more accurate detection. Multi-Scale Attention Fusion extracts features at different detail levels and prioritizes the most relevant ones. Cross-Level Feature Interaction allows network layers to share information, preserving fine details and global structure. Adaptive Fusion Strategy ensures clear and consistent fused data using semantic and structural guidance. MSAF-Net outperformed existing state-of-the-art models across four major datasets. It achieved 91.54% accuracy on UCF101 and 92.14% on ActivityNet. The model remains robust even if one sensor fails and can be pruned for faster performance on low-power hardware. [7]

N. Li, et al. UniFusOD is an end-to-end framework for infrared–visible image fusion and object detection. It combines visible images (rich texture and color) with infrared images (strong edges in low light). The system produces a single fused image optimized for clear visualization and accurate detection. FRA Mechanism uses multiple convolution kernels to capture both fine details and larger semantic features. UnityGrad Optimization balances fusion and detection gradients, preventing conflicts so both tasks train effectively. UniFusOD outperformed several state-of-the-art models on multiple datasets. Better Detection: Improved mAP by 0.8 (DroneVehicle) and 1.9 (M3FD). Superior Fusion: Produced images with higher information content and better structure. Robust Performance: Particularly effective at detecting small targets in complex environments. [8]

C. Konar, et al. This paper introduces a portable, edge-optimized surveillance system designed to detect modern security threats like hidden weapons and drones. By leveraging the Florence v2 Vision-Language Transformer, it aims to provide militarygrade object detection in resource constrained environments like remote borders or civilian infrastructure. Researchers adapted the Florence v2 model using Parameter Efficient Fine-Tuning (PEFT) and LowRank Adaptation (LoRA). The system was trained on 12,000 images, including GAN-generated synthetic data, and deployed on a Raspberry Pi 5 with an ESP32-S2 module for wireless alerts via the ESP-NOW protocol. The final model achieved 88.01% accuracy, outperforming YOLOv8 and FRCNN. It attained a mAP_{50} score of 64.83 while maintaining high efficiency on edge hardware, utilizing only 38.4% of the Raspberry Pi's RAM and 50.9% of its CPU power per 16.6ms frame. [9]

2.2 Comparison Table

Table 2.1: Comparison Table of Research Papers

Sl. No.	Research Paper	Learning Outcome
1	**Title:** Military target detection method based on EfficientDet and Generative Adversarial Network **Authors:** X. Zhuang, D. Li, Y. Wang, and K. Li **Published:** Engineering Applications of Artificial Intelligence, vol. 132, Jun. 2024	**Objective:** To detect targets in low visibility, noisy, or data-scarce combat zones **Methodology:** EfficientDet-D0 + Dual-Discriminator GAN **Core Idea:** Uses GANs to generate synthetic military training data and BiFPN for feature fusion
2	**Title:** An Efficient and Lightweight Deep Learning Model for Human Activity Recognition Using Smartphones **Authors:** Ankita, S. Rani, H. Babbar, S. Coleman, A. Singh, and H. M. Aljahdali **Published:** Sensors, vol. 21, Jun. 2021	**Objective:** To develop an efficient and lightweight model for identifying human activities using smartphone-based sensor data **Methodology:** CNN + Long Short-Term Memory (LSTM) **Core Idea:** Integrating spatial and temporal deep learning architectures to automate feature extraction from raw sensor data
3	**Title:** Detection and Classification of Different Weapon Types Using Deep Learning **Authors:** V. Kaya, S. Tuncer, and A. Baran **Published:** Applied Sciences, vol. 11, Aug. 2021	**Objective:** To propose a model for the automatic detection and classification of distinct types of handheld weapons **Methodology:** CNN based on VGGNet architecture **Core Idea:** Utilizing reduced layer complexity and highprecision CNNs to enable weapon identification on low-cost computing systems

Sl. No.	Research Paper	Learning Outcome
4	**Title:** Real-Time Versatile Object Detection With High Accuracy And Reduced Information Loss During Wartime **Authors:** S. D N and M. Dhanalakshmi **Published:** IEEE Conference, Sep. 2024	**Objective:** To detect unseen or new battlefield threats without manual re-labeling **Methodology:** CLIP Encoder + Modified C2f-Darknet **Core Idea:** Uses pseudo-boundary labeling for zero-shot detection and combines vision and language
5	**Title:** EDNet: Edge-Optimized Small Target Detection in UAV Imagery **Authors:** Z. Song, Y. Zhang, and A. A. R. M. A. Ebayyeh **Published:** IEEE SWC, Dec. 2024	**Objective:** To enable high-speed detection of tiny objects on low-power drone hardware **Methodology:** YOLOv10 + Faster Context Attention (C2fFCA) **Core Idea:** Uses XSmall detection layer for small object detection with hardware acceleration optimization
6	**Title:** Multi-modal action recognition via advanced image fusion techniques **Authors:** Z. Shou and D. Zhu **Published:** Frontiers in Physics, Aug. 2025	**Objective:** To recognize human/combatant actions accurately even if one sensor fails **Methodology:** MSAF-Net (Multi-Scale Attention-Guided Fusion) **Core Idea:** Dynamically prioritizes sensor data and switches to backup sensors when needed
7	**Title:** Infrared-Visible Image Fusion Meets Object Detection **Authors:** X. Xiang et al. **Published:** Remote Sensing, vol. 17, Nov. 2025	**Objective:** To solve gradient conflicts when training fusion and detection together **Methodology:** UniFusOD **Core Idea:** Use Fine-Grained Region Attention (FRA) mechanism and unified framework to fuse RGB and infrared images for accurate detection

Sl. No.	Research Paper	Learning Outcome
8	**Title:** PDGV-DETR: Object Detection for Secure On-Site Weapon and Personnel Location **Authors:** N. Li et al. **Published:** Sensors, vol. 26, Feb. 2026	**Objective:** To precisely detect and position weapons and personnel in static surveillance images while overcoming occlusion and complex background interference **Methodology:** PDGV-DETR framework **Core Idea:** Implementing a transformerbased end-to-end detection system that uses cross-scale semantic fusion to bridge the gap between high-level semantics and low-level image details
9	**Title:** Edge-Optimized Threat Detection with Florence v2 **Authors:** C. Konar, D. Das, D. S., K. Jc, and S. Singh **Published:** PeerJ Computer Science, vol. 12, Mar. 2026	**Objective:** To create a portable, edgeoptimized surveillance system capable of detecting military-relevant threat classes in resourceconstrained environments **Methodology:** Florence v2 VisionLanguage Transformer **Core Idea:** Leveraging multimodal vision-language models on compact edge hardware to provide military-grade situational awareness without relying on cloud infrastructure

The table 2.1 represents the comparison between the different research papers used for the literature review.

Chapter 3

REQUIREMENTS AND ANALYSIS

The requirements analysis for the VeerDrishti combat vision system encompasses a comprehensive evaluation of functional and non-functional requirements, ensuring that the system meets the operational needs of modern warfare while adhering to technical constraints. The analysis is structured to identify key features, performance metrics, and user interactions necessary for the successful deployment of the system in dynamic battlefield environments.

3.1 Functional Requirements

The functional requirements define the core capabilities of the project, including:

i. **Real-Time Object Detection:** The system must be capable of detecting and classifying combatants, vehicles, and weapons in real time using the integrated camera feed.

ii. **Visual Highlighting:** Detected threats should be visually highlighted on the display interface, providing clear and immediate feedback to the user.

iii. **Open Vocabulary Recognition:** The system should be able to identify novel or previously unseen threats using natural language supervision and language-vision encoders.

iv. **User Interface:** The system must provide an intuitive user interface that allows easy interpretation of the visual information to make informed decisions without distraction.

v. **Data Logging:** The system must provide secure local storage of detection events and video metadata for post-mission analysis and system auditing.

3.2 Non-Functional Requirements

The non-functional requirements address the performance, reliability, and usability aspects of the project, including:

i. **Low Latency:** The system must operate with minimal latency to ensure timely threat detection and response in high-stress combat scenarios.

ii. **Robustness:** The system should maintain high detection accuracy under varying environmental conditions, such as low light, occlusion, and dynamic backgrounds.

iii. **Scalability:** The system should be designed to accommodate future enhancements, such as additional threat categories or integration with other battlefield systems.

iv. **Security:** The system must ensure the confidentiality and integrity of the data collected, preventing unauthorized access and ensuring compliance with military data handling standards.

v. **Usability:** The system should be user-friendly, requiring minimal training for effective operation, and should not contribute to cognitive overload for the user.

3.3 Hardware Requirements

i. **Development Workstation:** High-performance laptop equipped with a dedicated GPU for code development, dataset preprocessing, and initial model debugging.

ii. **Edge Deployment Device:** High-end smartphone with an integrated Neural Processing Unit (NPU) to host the prototype application and execute real-time local inference.

iii. **Cloud Training Infrastructure:** Scalable cloud-based GPU instances utilized for training complex deep learning architectures and performing large-scale hyperparameter optimization.

3.4 Software Requirements

i. **Operating System:** Linux distribution like Ubuntu 26.04 LTS with stable CUDA/ROS support.

ii. **Programming Languages:** Python 3.9+ for model development, Rust for high-performance edge execution and Kotlin for mobile application development.

iii. **IDE/Tools:** VS Code, Jupyter Notebooks for experimentation, and Git for version control.

iv. **Core Framework:** PyTorch 2.0+ or TensorFlow 2.x (primarily PyTorch for YOLOv8/v10 implementation).

v. **Cloud Platform:** Google Cloud Vertex AI or AWS EC2 for large-scale model training.

Chapter 4

SYSTEM DESIGN

4.1 System Architecture

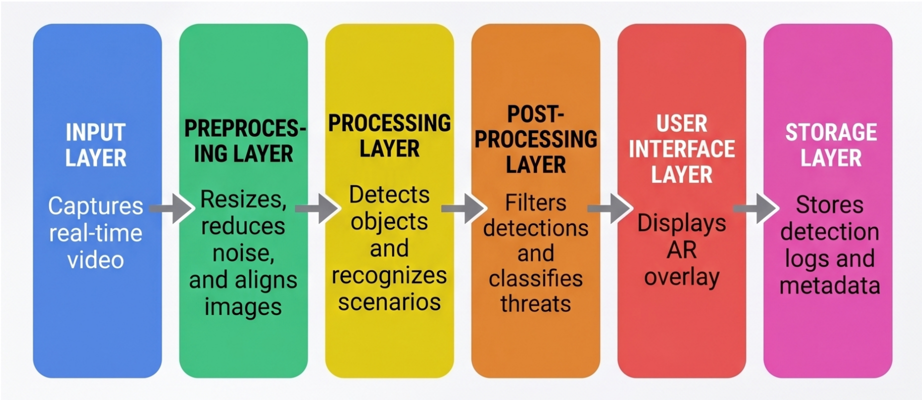

Figure 4.1: System Architecture

The figure 4.1 represents the tenetative system architecture of the VeerDrishti project.

The system architecture for the project is structured into six distinct layers that facilitate a smooth data flow from capture to storage. It begins with the Input Layer, which is responsible for capturing real-time video footage. This raw data is then passed to the Preprocessing Layer, where images are resized, aligned, and noise is reduced to ensure clarity for analysis. The Processing Layer serves as the core intelligence hub, detecting objects and recognizing specific scenarios, followed by the Post-Processing Layer, which filters these detections and classifies potential threats. Finally, the User Interface Layer provides an interactive experience by displaying an AR overlay of the results, while the Storage Layer logs all detection data and metadata for future reference.

4.2 System Workflow

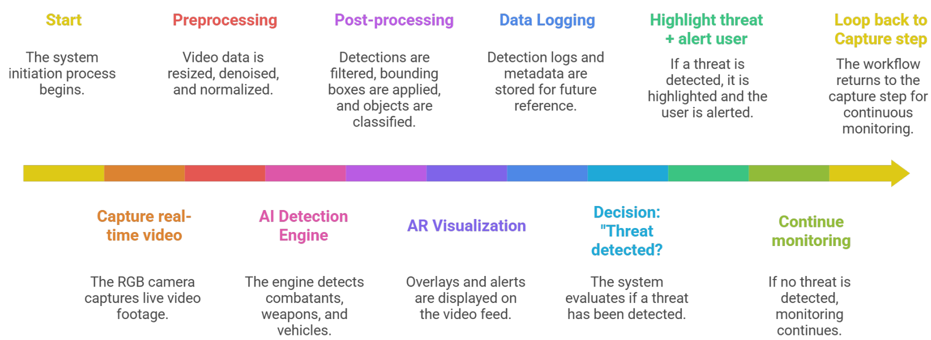

Figure 4.2: System Workflow

The figure 4.2 represents the tenetative system workflow of the VeerDrishti project.

The workflow follows a continuous loop designed for real-time monitoring and rapid response. Upon initiation, the system captures live RGB video and immediately subjects it to preprocessing steps like denoising and normalization. The AI Detection Engine then analyzes the feed to identify combatants, weapons, or vehicles. This leads to a critical decision point: if a threat is detected, the system applies bounding boxes, highlights the danger via AR visualization, alerts the user, and logs the data. If no threat is found, the system simply continues its monitoring cycle, returning to the capture step to ensure persistent surveillance.

Chapter 5

SCHEDULING

5.1 Gantt Chart

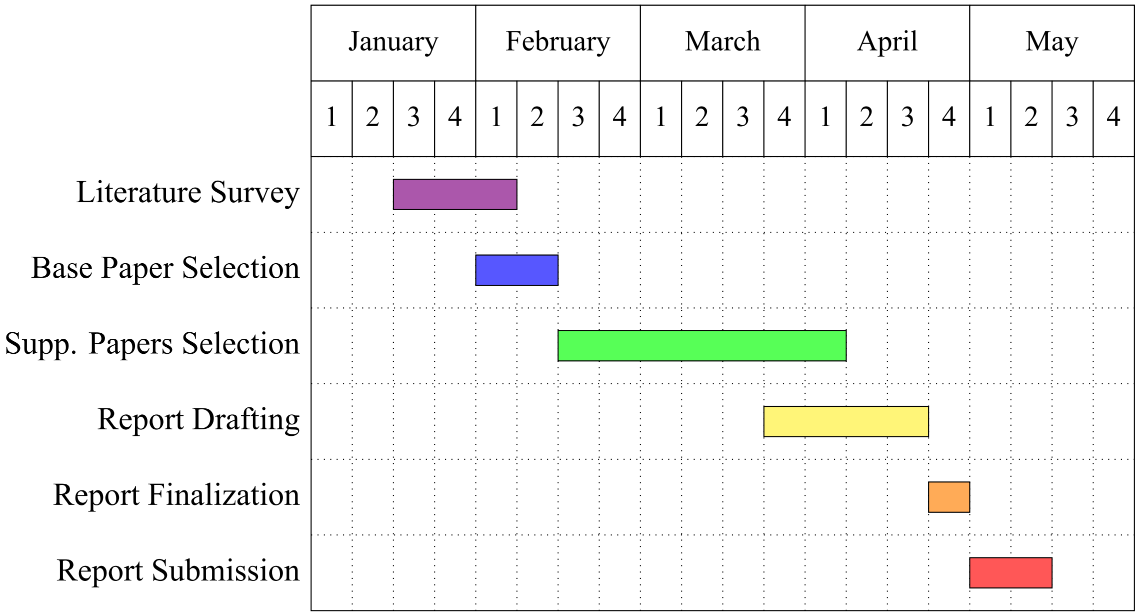

Figure 5.1: Gantt Chart for Major Project (Phase-1)

The figure 5.1 represents the timeline for the Major Project (Phase-1).

The project follows a structured timeline spanning from January to May. The initial phase begins in the second week of January with a three-week Literature Survey, which concludes in early February. This is immediately followed by the Base Paper Selection, spanning the first two weeks of February. Supporting Papers Selection constitutes the longest research block, starting in late February and continuing through the end of April. As research nears completion, Report Drafting commences in early April and runs for three weeks. The final stages are tightly scheduled in May, with Report Finalization occurring in the first week, followed by the final Report Submission during the second and third weeks of the month. This sequential approach ensures a rigorous foundation of research before the final documentation is prepared and delivered.

5.2 Month-wise Progress

The project timeline spans from January to May 2026, transitioning from initial literature research to final documentation. The month-wise progress details are as follows:

Table 5.1: Comparison Table of Research Papers

Sl. No.	Month	Progress Details
1	January	Conducted a comprehensive literature survey to identify relevant research papers and technologies in the domain of real-time combat vision systems. Identified research papers that could serve as the base paper and foundation for our project.
2	February	Selected the base paper and conducted extensive reading and analysis to understand the current state of the art, identify gaps in existing solutions, and select a base paper that aligns with our project objectives.
3	March	Selected 10 supplementary papers that provide additional insights, methodologies, and technologies relevant to our project. These papers contribute to the development of our combat vision system by providing a broader understanding of the field and potential approaches to implementation. Began intial work to draft the project report.
4	April	Drafted the project report, synthesizing information from the base paper and supplementary papers. Structured the report to clearly articulate the project details and specifications. Iteratively refined the report based on feedback and additional insights gained during the drafting process and drafted the final project report.
5	May	Submitted the finalized project report, ensuring that all sections are complete, well-organized, and adhere to the required guidelines.

The table 5.1 represents the month-wise progress of the Major Project (Phase-1).

Chapter 6

CONCLUSION

The project advances tactical situational awareness by combining state-of-the-art AI with wearable combat technology. It employs the lightweight YOLOv8 architecture and multimodal image fusion to enable real-time, autonomous threat detection in high-stress environments. By integrating visible and thermal imaging, the system remains effective in challenging conditions such as darkness, smoke, and low visibility. This reduces cognitive load on personnel by converting complex visual data into actionable insights, enabling faster and more accurate decision-making. The system's robustness is enhanced through GAN-based synthetic data augmentation, which addresses limited military datasets, and CLIP-based open-vocabulary detection, allowing adaptation to unseen threats via natural language guidance. Additionally, its edge-optimized design enables high-speed, low-latency inference on portable, smartphone-based hardware. This ensures reliable performance even in signal-denied environments while maintaining data security and operational efficiency, ultimately supporting rapid and informed responses in critical combat scenarios.

Looking forward, the potential for VeerDrishti extends into a scalable framework for future defense innovations. Future iterations could integrate more complex action recognition models to predict hostile intent or incorporate advanced feature-fusion strategies to further enhance the detection of micro-scale targets. In conclusion, VeerDrishti offers a versatile and reliable solution for modern national security. By harmonizing cutting-edge computer vision research with the practical, high-stakes requirements of the frontline, the project provides a definitive technological advantage, ensuring enhanced safety and operational efficiency for ground units in the evolving landscape of modern warfare.

REFERENCES

[1] X. Zhuang, D. Li, Y. Wang, and K. Li, "Military target detection method based on EfficientDet and Generative Adversarial Network," *Engineering Applications of Artificial Intelligence*, vol. 132, Jun. 2024, Art. no. 107896. DOI: `10.1016/j.engappai.2024.107896`

[2] Ankita, S. Rani, H. Babbar, S. Coleman, A. Singh, and H. M. Aljahdali, "An efficient and lightweight deep learning model for human activity recognition using smartphones," *Sensors*, vol. 21, no. 11, Jun. 2021, Art. no. 3845. DOI: `10.3390/s21113845`

[3] V. Kaya, S. Tuncer, and A. Baran, "Detection and classification of different weapon types using deep learning," *Applied Sciences*, vol. 11, no. 16, Aug. 2021, Art. no. 7535. DOI: `10.3390/app11167535`

[4] S. D N and M. Dhanalakshmi, "Real-time versatile object detection with high accuracy and reduced information loss during wartime," in *2024 International Conference on Integration of Emerging Technologies for the Digital World (ICIETDW)*, IEEE, Sep. 2024, pp. 1–7. DOI: `10.1109/ICIETDW61607.2024.10939768`

[5] Z. Song, Y. Zhang, and A. A. R. M. A. Ebayyeh, "EDNet: Edge-optimized small target detection in UAV imagery - faster context attention, better feature fusion, and hardware acceleration," in *2024 IEEE Smart World Congress (SWC)*, IEEE, Dec. 2024, pp. 829–838. DOI: `10.1109/SWC62898.2024.00141`

[6] Z. Shou and D. Zhu, "Multi-modal action recognition via advanced image fusion techniques for cyber-physical systems," *Frontiers in Physics*, vol. 13, Aug. 2025, Art. no. 1576591. DOI: `10.3389/fphy.2025.1576591`

[7] X. Xiang et al., "Infrared-visible image fusion meets object detection: Towards unified optimization for multimodal perception," *Remote Sensing*, vol. 17, no. 21, Nov. 2025, Art. no. 3637. DOI: `10.3390/rs17213637`

[8] N. Li et al., "PDGV-DETR: Object detection for secure on-site weapon and personnel location based on dynamic convolution and cross-scale semantic fusion," *Sensors*, vol. 26, no. 5, Feb. 2026, Art. no. 1542. DOI: `10.3390/s26051542`

[9] C. Konar, D. Das, D. S., K. Jc, and S. Singh, "Edge-optimized threat detection with Florence v2 for autonomous surveillance in resource-constrained environments," *PeerJ Computer Science*, vol. 12, Mar. 2026, Art. no. 3579. DOI: `10.7717/peerj-cs.3579`